P11

A few logs suffice to build (almost) all trees (I)

Authors P�ter L. Erd?s, Michael A. Steel, L�szl� A. Sz�kely, Tandy J. Warnow

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Abstract

A phylogenetic tree, also called an “evolutionary tree,” is a leaf-labeled tree which represents the evolutionary history for a set of species, and the construction of such trees is a fundamental problem in biology. Here we address the issue of how many sequence sites are required in order to recover the tree with high probability when the sites evolve under standard Markov-style i.i.d. mutation models. We provide analytic upper and lower bounds for the required sequence length, by developing a new polynomial time algorithm. In particular, we show when the mutation probabilities are bounded the required sequence length can grow surprisingly slowly (a power of log n) in the number n of sequences, for almost all trees.

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Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What problem does the paper try to solve? Write it in one sentence, in your own words.
2. What is unknown or hidden, and what can actually be observed or measured?
3. What method do the authors use? Is it new, borrowed from another paper, or a modification of one?
4. What evidence do they give (proofs, simulations, experiments, real data)?
5. What assumptions or limitations can you find?
6. Which other papers in the corpus does it cite, use, extend or resemble? Note the page or section.